Spend the retention budget where it returns the most

Profit-aware churn and retention targeting - Business Analytics capstone, Topic 7

- Decision: which customers get a retention offer, which offer, and how much to spend
- Answer: rank by expected value, not by churn risk alone, and prove the save rate in a pilot
- Ask: approve one randomised pilot wave of 1,620 customers

Target by expected value, then prove the save rate

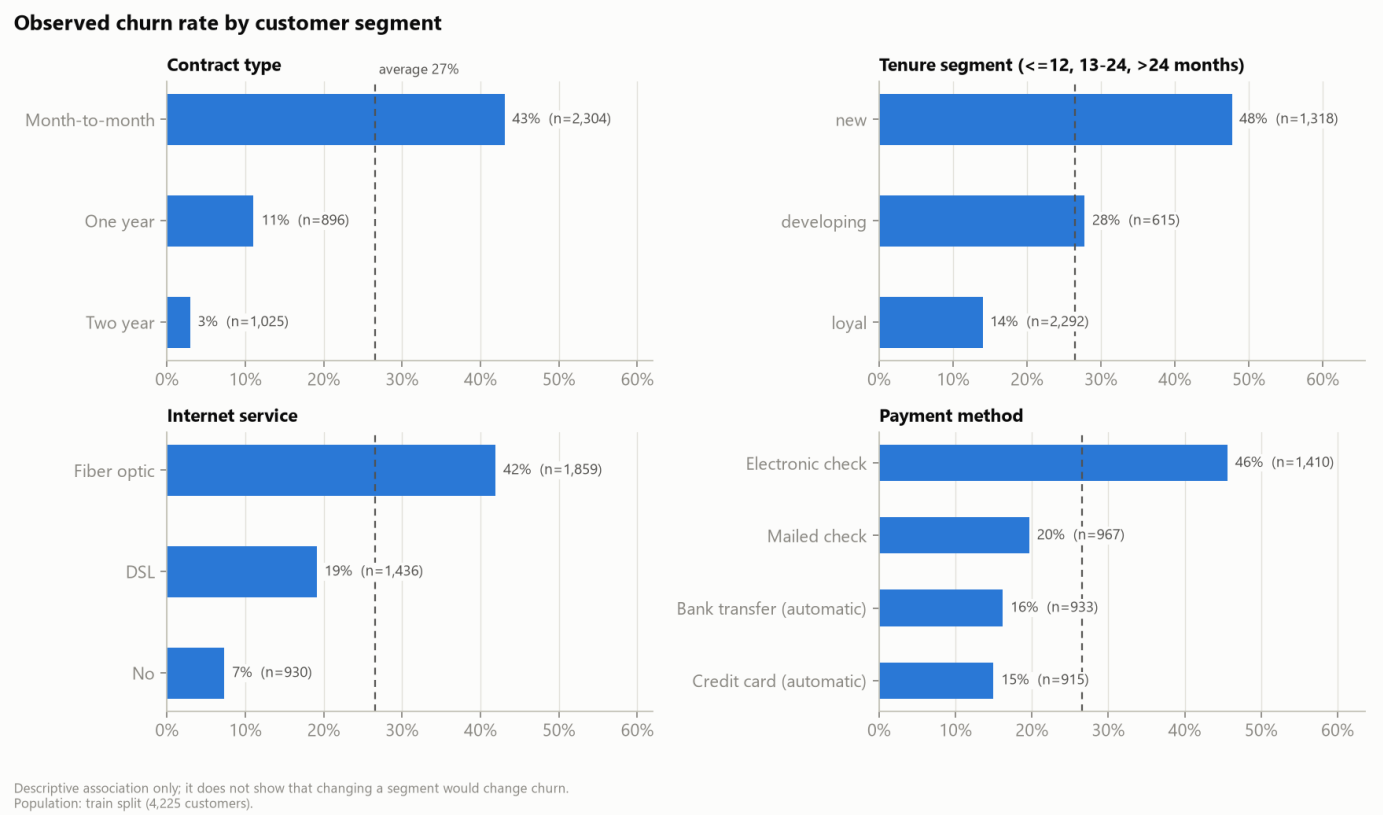
Executive summary

- Same $3,500 budget: +$1,368 expected versus calling the riskiest customers
- Who: fibre customers on monthly contracts with high bills; not low-risk or low-bill customers
- Budget: about $45,737 for the base; stop where the next step returns under 10%
- Who to target is robust; how much an offer saves (20% assumed) is not
- Ask: one pilot wave, half held out, about $22,875 of offers and contacts

Churn puts $468,468 of margin at risk

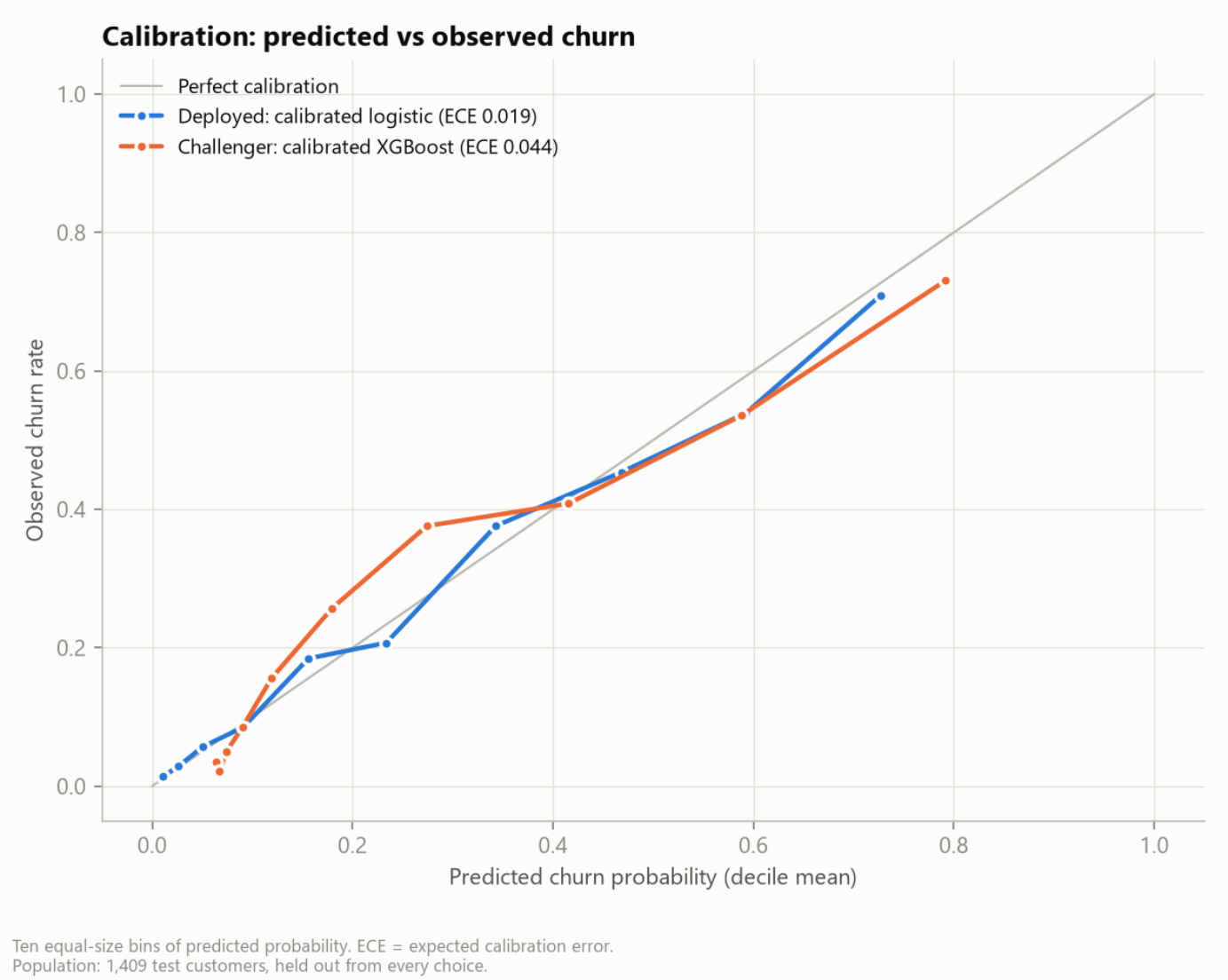
Today's rule calls the riskiest customers, whatever they are worth

- 26.5% of customers churn; monthly-contract, new, fibre and e-cheque customers most
- The budget covers an offer for 10% of customers
- Every offer costs money, even when the customer would have stayed
- Owner: Head of Retention; Finance approves budget; Legal/DPO approves contact



The model ranks churners well and is checked in money

- Logistic regression, chosen over random forest and XGBoost by a rule fixed in advance
- Finds churners: ROC-AUC 0.83; PR-AUC 0.62 vs 0.27 by chance
- Probabilities within 1.9 points of observed churn
- In money the three models are within $194 on validation
- It predicts who leaves, not who an offer would keep



Ten equal-size bins of predicted probability. ECE = expected calibration error.
Population: 1,409 test customers, held out from every choice.

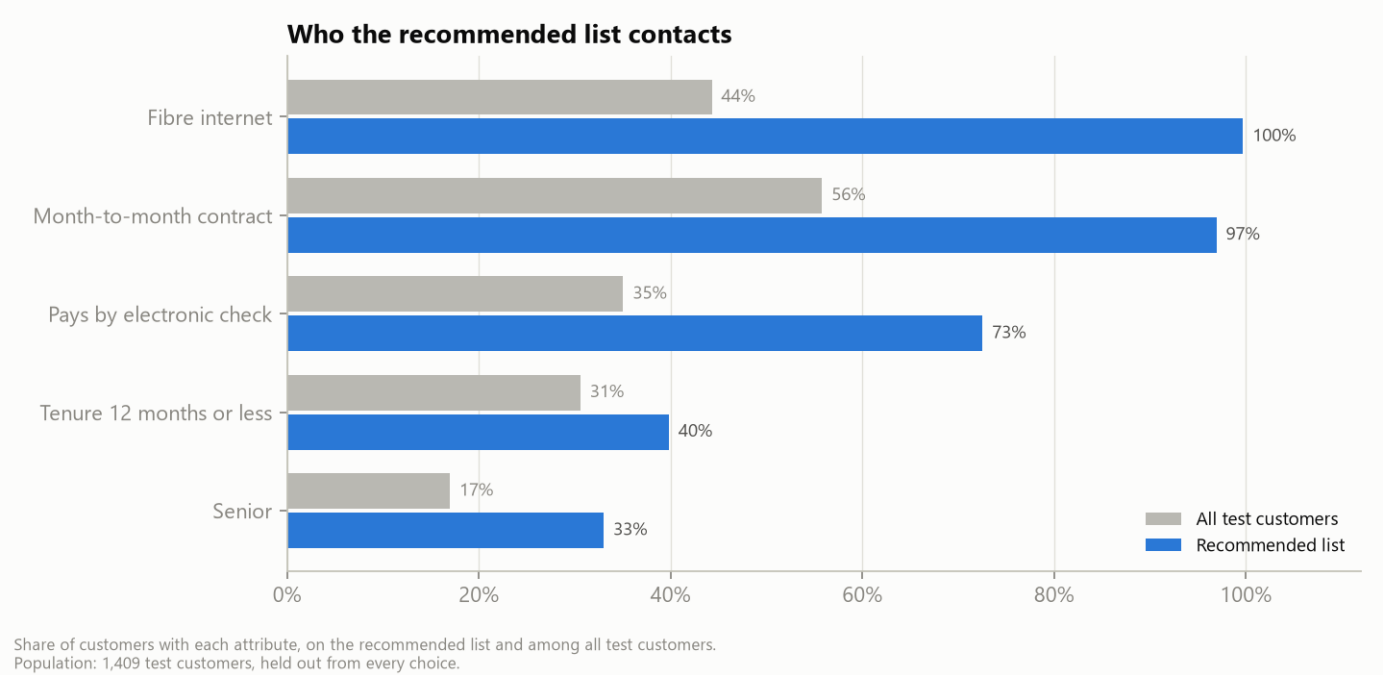
Offer only where expected value beats the cost

Expected value = churn probability x save rate x customer value - (offer + contact cost)

- Contact only customers with positive expected value
- An optimiser picks customers and one offer each within budget, at most 30% contacted
- Four offers with eligibility rules: bill credit, contract upgrade, service bundle, loyalty reward
- Every cost and save rate is a visible, sourced assumption

The list is fibre, month-to-month and high-bill

- 100% fibre and 97% monthly contracts, against 44% and 56% overall
- Median bill $89 against $70
- No offer for low-risk customers: most would stay anyway
- Few offers for risky low-bill customers: their value cannot repay the offer



Share of customers with each attribute, on the recommended list and among all test customers.
Population: 1,409 test customers, held out from every choice.

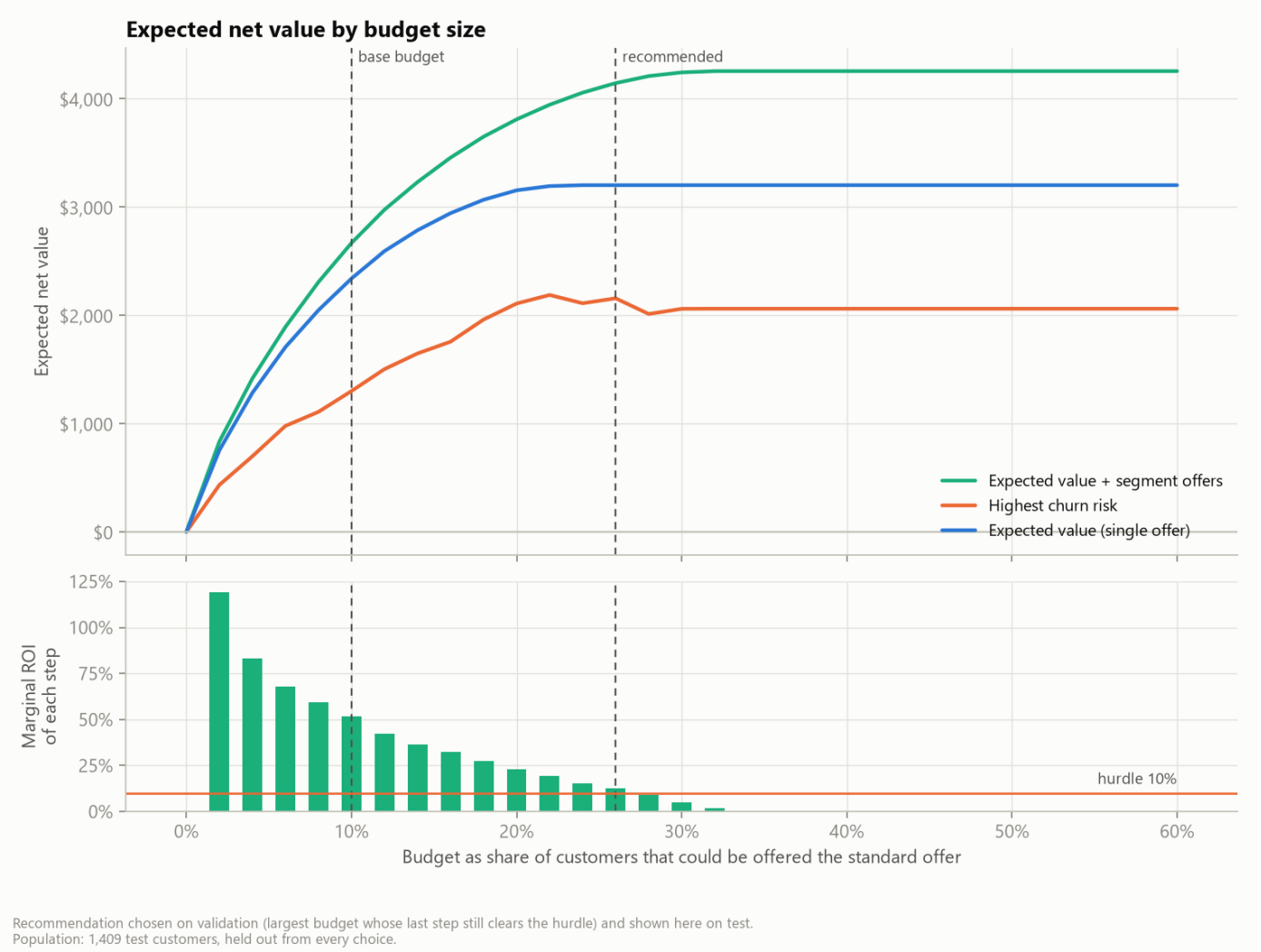
Same budget, more value

- +$1,368 expected: $1,039 from who we target, $328 from which offer

Rule	Customers	Net value	ROI	Precision	HR/HV coverage
Highest churn risk (today)	140	$1,301	37%	71%	12%
Expected value, one offer	140	$2,340	67%	60%	96%
Recommended: + segment offers	153	$2,668	76%	60%	98%

Spend about $45,737 - then stop

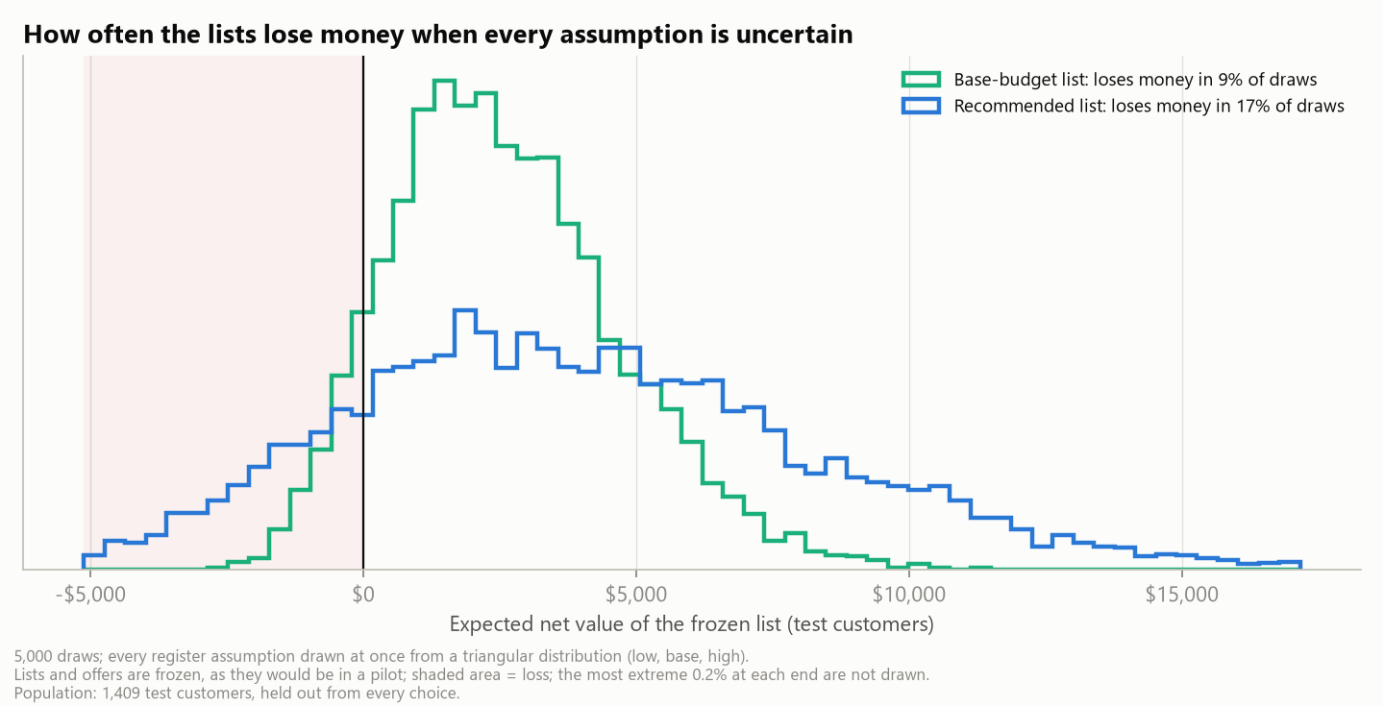
- Each extra step reaches customers with less expected value
- Keep steps that return at least 10% (set on validation): $6.49 per customer
- Test customers: 324 offers, $9,150, net $4,145 (ROI 45%)
- Beyond 26% of customers nobody is worth an offer



Recommendation chosen on validation (largest budget whose last step still clears the hurdle) and shown here on test.
Population: 1,409 test customers, held out from every choice.

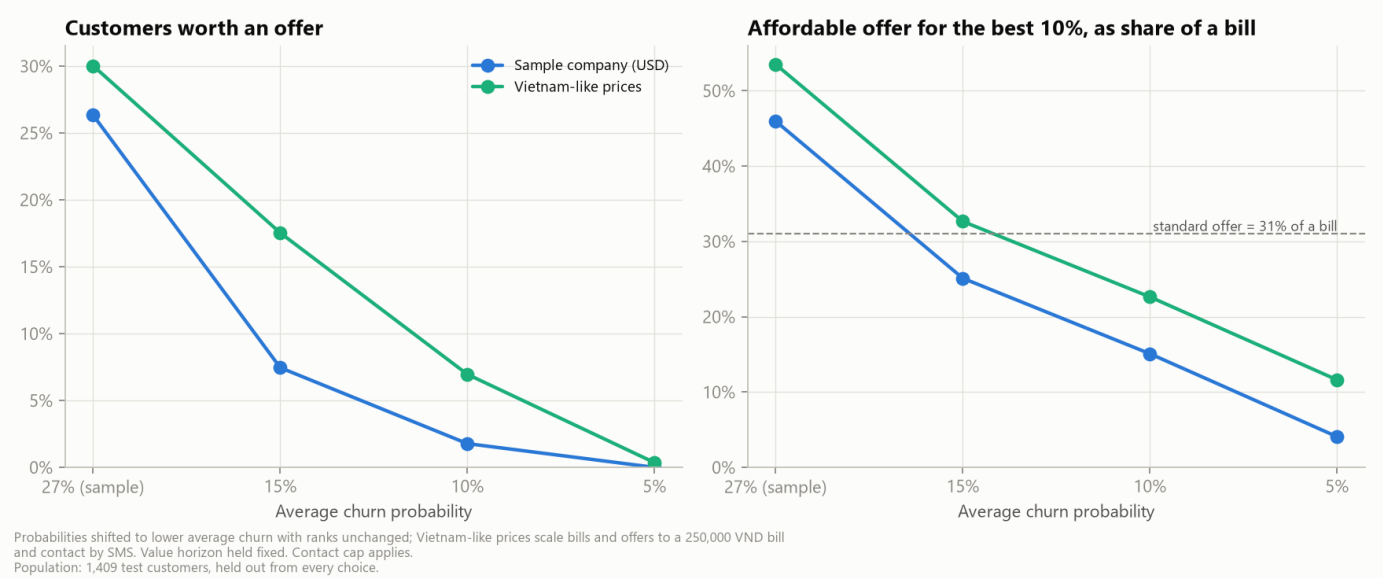
Who to target is robust; how much an offer saves is not

- Break-even save rate: 11.3% (base list), 13.8% (recommended); assumed 20%
- All assumptions uncertain: the recommended list loses money in 17% of draws
- Beats risk-only in 100% of draws
- Only 46% of churners left for price or a competitor's offer



Outside the sample the margins get thinner

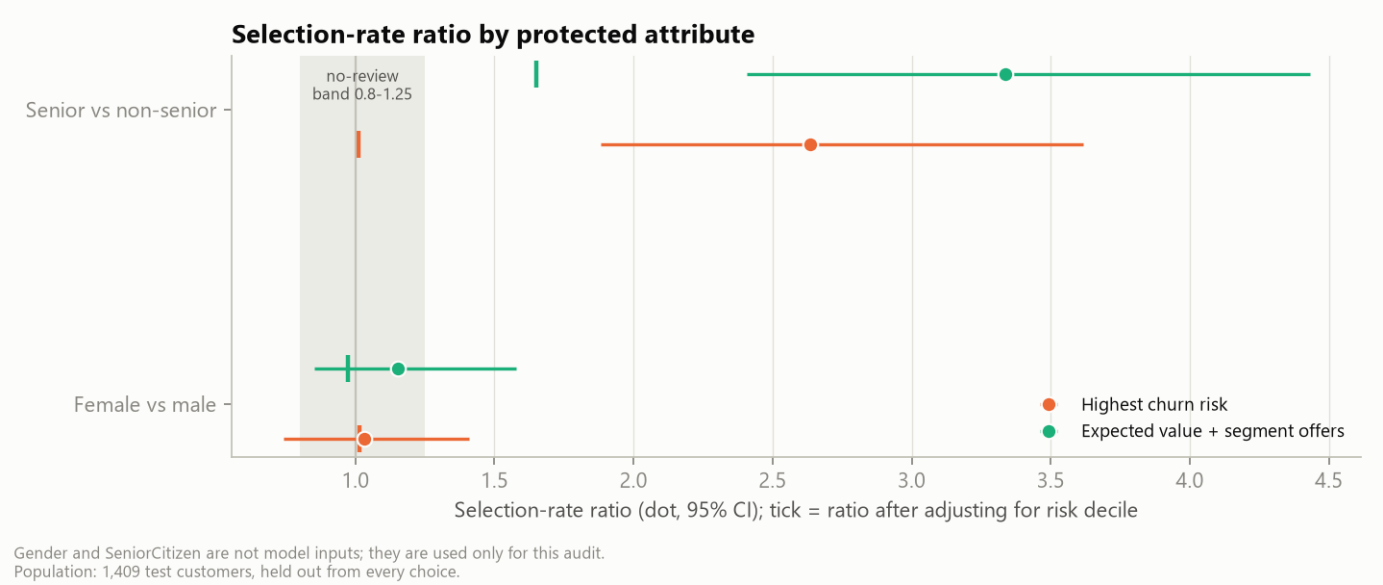
- Real churn is far below the sample's 27%
- At 15% churn only 7% of customers are worth this offer
- Vietnamese prices: small bills but SMS contact keeps 18% worth an offer
- Lower churn needs larger pilots; rebuild the model on our own data



Probabilities shifted to lower average churn with ranks unchanged; Vietnam-like prices scale bills and offers to a 250,000 VND bill and contact by SMS. Value horizon held fixed. Contact cap applies.
Population: 1,409 test customers, held out from every choice.

Fairness, experience and compliance are managed

- Seniors contacted 3.3x as often as others (2.6x under today's rule)
- 1.7x after adjusting for risk: reviewed with Legal/DPO before each wave
- Age and gender are not model inputs; every wave is audited
- Consent, Do-Not-Call checks, contact limits and a data-protection impact assessment



Gender and SeniorCitizen are not model inputs; they are used only for this audit.
Population: 1,409 test customers, held out from every choice.

The pilot measures what we assumed

- 1,620 customers: 810 offered, 810 held out at random
- Save rate = 1 - churn of offered / churn of held out, after 90 days
- One wave detects a save rate of about 12%; break-even is 13.8%
- Cost about $22,875; expected net value of the offered half $10,363
- Decision about 19 weeks after approval: set-up, outcome window, read-out

Decisions requested

- Approve one pilot wave: 1,620 customers, half held out, about $22,875
- Name owners: Retention, Call centre, Finance, Legal/DPO
- Agree the rule now: go if significant and at least 13.8%; stop if clearly below
- Otherwise extend, at most 2 more waves

Live demo: the dashboard

- Decision page: KPIs, who gets an offer, action list and export
- Assumptions page: change the save rate and watch the chance of a loss move
- Model evidence: odds ratios and the points scorecard